Weekly Progress Report — Week 5

Project Title:

AI Business Intelligence Dashboard

Student/Developer:

Filza Rehman

Track / Difficulty:

AI & Machine Learning / Advanced

Live Production URL:

https://ai-business-intelligence-dashboard-dq7pd8yxpbusnzkrrsvx9y.streamlit.app/

GitHub Repository:

https://github.com/filzarehman/AI-Business-Intelligence-Dashboard

1. Project Summary & Objectives

- Built an end-to-end business intelligence platform tailored for small-to-medium e-commerce enterprises to consolidate fragmented operational data.
- Integrated automated KPI tracking, rule-based diagnostic summaries, predictive linear regression forecasting (30-day horizon), and server-side executive PDF report generation.

2. Key Accomplishments

- Implemented multidimensional data filtering across dynamic date ranges.
- Designed modular analytical interfaces using Plotly (Revenue Trends, Sales & Conversion, Marketing Attributions).
- Integrated machine learning time-series regression to project future revenue trajectories.
- Integrated fpdf2 for one-click downloadable executive summaries.
- Deployed full-stack application to Streamlit Community Cloud.

Daily Work Log (6-Day Breakdown)

- **Day 1: Problem Definition & Data Schema**
  - Defined e-commerce problem statements and commercial target markets (USA, UK, UAE, Australia, Europe).
  - Structured synthetic dataset schema (ecommerce_data.csv) covering revenue, orders, traffic, and marketing spend.
- **Day 2: Core Data Pipelines & KPI Architecture**
  - Developed src/data_processing.py to calculate core metrics (AOV, conversion rate, period-over-period delta).
  - Built initial modular project structure and environment configurations.
- **Day 3: Visual Interface & Plotly Integration**
  - Designed black-and-blue dark mode UI styling inside app.py.
  - Implemented responsive Plotly charts for Revenue, Sales, and Marketing tabs.
- **Day 4: AI Insights & Machine Learning Forecasting**

    - Built src/forecasting.py using Scikit-learn Linear Regression to generate 30-day forward forecasts.
    - Created heuristic business diagnostic engine in src/ai_insights.py to produce plain-English recommendations.
- **Day 5: PDF Export Engine & Cloud Deployment**
    - Integrated fpdf2 buffer export for dynamic executive PDF generation.
    - Configured dependencies in requirements.txt and completed production deployment to Streamlit Community Cloud.
- **Day 6: Documentation, Auditing & Final Submission**
    - Completed full GitHub README.md with system architecture and commercial viability details.
    - Verified calculation parity across raw data tables and chart outputs.

Problems Encountered & Technical Solutions

- **Problem 1: Streamlit State Reset on Widget Interaction**
    - *Challenge:* Date filter changes caused full script reruns, occasionally misaligning period-over-period delta calculations.
    - *Solution:* Modularized calculation helper functions (calculate_period_changes) to compute slices dynamically based on the active filtered dataframe.
- **Problem 2: PDF Export File Buffer in Web Deployment**
    - *Challenge:* Direct filesystem writes caused deployment permission warnings on headless cloud instances.
    - *Solution:* Implemented in-memory streaming via Python's io.BytesIO() buffer, allowing fpdf2 binary output to serve directly through Streamlit's st.download_button.
- **Problem 3: Git Remote Divergence during Sync**
    - *Challenge:* Fast-forward push rejections caused by remote branch modifications.
    - *Solution:* Resolved local-remote merge states using Git fetch, hard resets to remote origin, and clean branch rebasing.

Next Week's Plan

- Integrate persistent database connections (e.g., PostgreSQL / Snowflake) to replace static CSV uploads.
- Expand the insight layer by incorporating an external LLM endpoint for dynamic multi-turn data Q&A.
- Add user authentication and role-based client dashboard access.